Yalla: Estimating Muscle Growth, Maintenance, and Decline from Training Logs

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Abstract

Yalla estimates, per muscle and for the body overall, whether recent training is enough to grow, hold, or slowly lose muscle size. The estimate is a read on *training stimulus* inferred from logged sets, loads, and reps, not a measurement of the body. Two axes drive it: a *dose* axis (effort-weighted weekly sets against volume landmarks) and a *trend* axis (progressive overload measured as the better of a set trend and a load trend). Set effort is captured with a three-level control whose default is estimated from the load–rep relationship against the lifter's own best set. This document defines every quantity, the equations that combine them, the decision rules that produce a state, and the evidence each parameter rests on. It also states the model's assumptions, its validation status, and where it is deliberately conservative. The specification corresponds to the functions `growthStatus()`, `inferEffort()`, and `planForecast()` in `app.js`.

1 Model overview

The unit of analysis is a muscle group g (chest, lats, quads, and so on). Training history is a set of logged entries; each entry e records one exercise in one session with a date d_e, a top-set load w_e (kg), top-set reps r_e, a set count n_e, an effort code $f_e \in \{0, 1, 2\}$, and a stored volume-load v_e.

Each muscle is placed in one of three states, combined from the two axes: *growing* (adequate dose and progressing), *holding* (a maintenance dose, or flat), and *under-stimulated* (below the volume needed to hold size). The dose axis answers whether there is enough stimulus this week; the trend axis answers whether the stimulus is rising, flat, or falling. A muscle grows only when both are satisfied: enough sets and a rising trend. This mirrors the requirement that hypertrophy needs both sufficient volume and progressive overload.

2 Volume attribution

2.1 Fractional set counting

Each exercise contributes to its primary muscle at full weight and to each secondary muscle at half. Writing the attribution weight of exercise x to muscle g as $\alpha(x, g)$:

$$\alpha(x,g) = \begin{cases} 1 & \text{if } g \text{ is the primary mover of } x, \\ 0.5 & \text{if } g \text{ is a secondary mover of } x, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

The 0.5 weight for indirect work is the fractional counting scheme that best fit the hypertrophy data in the Pelland et al. dose–response meta-regression [5].

2.2 Effort weighting

Only sets taken reasonably close to failure drive much growth, so each set is scaled by an effort factor $\varepsilon(f)$ before it counts:

$$\varepsilon(f) = \begin{cases} 0.5 & \text{Easy } (f{=}0,\ \geq 4 \text{ reps in reserve}), \\ 1 & \text{Hard } (f{=}1,\ {\sim}1\text{–}3 \text{ in reserve; the default}), \\ 1 & \text{Max } (f{=}2,\ 0\text{–}1 \text{ in reserve}). \end{cases} \tag{2}$$

Max carries no growth premium over Hard: hypertrophy is comparable across roughly 0–4 reps in reserve, and training to failure mainly adds fatigue [7]. Max still helps the load trend indirectly through the extra reps it buys. Entries with no recorded effort (all logs predating the feature) default to Hard, so $\varepsilon = 1$ and historical counts are unchanged.

2.3 Weekly effective sets per muscle

Time is bucketed into $N = 10$ rolling weeks; week index $k = 9$ is the current week. The effective weekly sets for muscle g in week k sum every contributing entry:

$$S_{g,k} = \sum_{e \,\in\, \text{week } k} n_e \, \varepsilon(f_e) \, \alpha(x_e, g). \tag{3}$$

This same effort-weighted, fractional count feeds the weekly-sets chart, the balance radar, and the "hard sets" ring, so every surface in the app reads consistently.

3 Effort inference

Proximity to failure cannot be recovered from load and reps alone: the same 100 kg × 8 can be an all-out set or an easy one. So effort is a self-report. To keep friction to one optional tap, its default is estimated from the numbers and shown pre-selected; the lifter overrides only when the estimate is wrong.

3.1 Estimated one-rep max (anchor)

Set strength is summarised by the Epley one-rep-max estimate [2], with a rep denominator K (population default $K = 30$; Section 3.3 calibrates it per lifter):

$$\varphi(w, r) = w \left(1 + \frac{r}{K}\right). \tag{4}$$

The anchor for exercise x is the best estimate over the lifter's *prior* sessions for that exercise (the current, in-progress session is excluded):

$$A_x = \max_{e \,\in\, \text{hist}(x)} \varphi(w_e, r_e). \tag{5}$$

3.2 Reps in reserve and classification

Inverting (4) gives the reps a lifter should complete at load w before failure; subtracting the reps actually done estimates reps in reserve:

$$\text{RIR} = K \left(\frac{A_x}{w} - 1\right) - r, \tag{6}$$

evaluated on the session's hardest set for that exercise (the one with the largest φ). The estimate maps to an effort code:

$$f = \begin{cases} 2 \text{ (Max)} & \text{if RIR} \le 1, \\ 0 \text{ (Easy)} & \text{if RIR} \ge 4, \\ 1 \text{ (Hard)} & \text{otherwise.} \end{cases} \tag{7}$$

The anchor is recomputed every time rather than frozen after a calibration period. A rolling anchor adapts to deloads and off days and never locks in a wrong baseline the way a one-time calibration could. When there is no prior baseline for the lift, or the exercise carries no external load (bodyweight or timed holds), (6) is not evaluated and the default stays Hard.

3.3 Self-calibrated rep denominator

The denominator K in (4) varies by lifter and by lift. A set logged as Max is a near-true failure point, so two Max sets of the same exercise at different loads pin K: setting φ equal for both, $w_1(1 + r_1/K) = w_2(1 + r_2/K)$, which solves to

$$K = \frac{w_2 r_2 - w_1 r_1}{w_1 - w_2}. \tag{8}$$

Valid within-exercise pairs are pooled across the whole log, the median is taken and clamped to $[20, 40]$, and $K = 30$ is used until enough failure data exists. This grounds the effort estimate in the lifter's own data rather than a fixed population constant.

4 Volume landmarks

Four thresholds, in effective sets per muscle per week, define the dose axis. They follow the MEV/MAV framework and the dose–response meta-analyses.

Symbol (value)	Name	Meaning
MV (2)	Maintenance volume	Floor to hold existing size. Below it, a previously-trained muscle reads as under-stimulated.
MEV (6)	Minimum effective volume	Lower edge of the productive growth window; at or above it, an adequate dose is present.
target (10)	Practical target	Where most of the dose–response benefit has accrued; the balance-radar goal.
MAV (20)	Upper productive bound	Beyond it, added sets mostly add fatigue rather than growth.

A single set of landmarks is applied to every muscle. Real per-muscle landmarks differ, but one baseline keeps the output legible. The maintenance floor is deliberately low: built muscle is held on very little (as little as roughly one-ninth of the building volume in young trained adults [1]), so a 2-set floor avoids flagging loss too early.

4.1 Continuous backbone

The four landmarks discretise a continuous relationship [9, 5]. We model the fraction of the attainable per-muscle growth stimulus at s effective sets per week as a saturating curve $\rho(s)$, calibrated so the practical target captures about 80% of what is attainable:

$$\rho(s) = 1 - e^{-s/s_0}, \qquad s_0 = \frac{\text{target}}{\ln 5} \approx 6.21. \tag{9}$$

The landmarks are read off this single curve rather than chosen independently: $\rho(\text{MV}) \approx 0.28$, $\rho(\text{MEV}) \approx 0.62$, $\rho(\text{target}) \approx 0.80$, and $\rho(\text{MAV}) \approx 0.96$. So MEV is the ≈ 0.6 knee, target the 0.8 point, and MAV the ≈ 0.95 near-plateau. This replaces four independent thresholds with one meta-analytic function; the growth signal also reports $\rho(D_g)$ as a per-muscle share of the attainable growth stimulus.

5 Temporal windows

Both axes compare a recent window against an earlier one, each a three-week mean over the 10-week series. For any weekly series X_g:

$$\text{recent}(X_g) = \frac{1}{3}\sum_{k=7}^{9} X_{g,k}, \qquad \text{earlier}(X_g) = \frac{1}{3}\sum_{k=4}^{6} X_{g,k}. \tag{10}$$

The three-week mean smooths single missed or heavy sessions. The recent window is weeks 8–10 (the last three), the earlier window is weeks 5–7; the oldest three weeks provide history for charts but do not enter the comparison. Because the window is recent, the signal responds within about three weeks of a genuine change in training.

6 The two axes

6.1 Dose

The current dose for muscle g is the recent mean of effective weekly sets:

$$D_g = \text{recent}(S_g). \tag{11}$$

6.2 Trend: progressive overload two ways

Overload can come from more effective sets or from heavier load; either counts as progress [6, 8]. A set trend and a load trend are formed as recent-over-earlier ratios. The load series $L_{g,k}$ is the best estimated one-rep max reached in week k on exercises for which g is the primary mover:

$$T_g^S = \frac{\text{recent}(S_g)}{\text{earlier}(S_g)}, \qquad T_g^L = \frac{\text{recent}(L_g)}{\text{earlier}(L_g)}. \tag{12}$$

Each ratio is defined only when its earlier window is positive; otherwise it is treated as absent. The combined trend takes the better of the two:

$$\tau_g = \max\Big(T_g^S,\, T_g^L\Big), \tag{13}$$

with an absent ratio contributing 0 to the maximum, and τ_g itself absent only when both ratios are absent (a muscle just started, with no earlier baseline). Taking the maximum prevents classic strength progression (add load, drop reps) from being misread as a decline: raw tonnage would fall, but T^L rises.

An earlier version used tonnage, $V = w\,r\,n$, as the sole trend signal. Tonnage rewards total kilograms moved, so lighter high-rep work inflates it while heavier low-rep work deflates it; it is a poor proxy for stimulus, and (13) replaces it.

7 Per-muscle state

The state σ_g is a function of the dose D_g and the trend τ_g, evaluated top to bottom (first matching rule wins):

1. **Under-stimulated** if $D_g < \mathrm{MV}$ (below the maintenance floor of 2).
2. **Under-stimulated** if τ_g exists and $\tau_g < 0.75$ (training eased off sharply).
3. Otherwise, if $D_g \geq \mathrm{MEV}$ (adequate dose, ≥ 6): **growing** if τ_g is absent or $\tau_g \geq 1.08$; **holding** otherwise ($0.75 \leq \tau_g < 1.08$).
4. Otherwise (maintenance band, $2 \leq D_g < 6$): **growing** if $\tau_g \geq 1.15$ and $D_g \geq 5$ (climbing toward a growth dose); **holding** otherwise.

The asymmetry is intentional: a muscle grows only with both an adequate dose and a rising trend, but it can be flagged under-stimulated by a low dose alone. Holding is the wide middle band: enough volume to keep size, below the dose that adds it.

7.1 Borderline states

The inputs carry known noise: effort-weighted sets to about ± 1 set, and each trend ratio to about ± 0.06. When a muscle's dose or trend sits within that range of a decision boundary, it is flagged *borderline* and shown with a tilde instead of a crisp verdict, so a near-miss is not read as certainty. The sensitivity analysis (Section 11) shows the growth-trend boundary is where this matters most.

8 Overall verdict

Let n_{grow}, n_{hold}, n_{shrink} count trained muscles in each state, and m be the number of trained muscles. Let τ_{all} be the recent-over-earlier ratio of whole-body weekly volume. The headline is:

1. **Need more data** if fewer than 3 weeks show any training, or $m = 0$.
2. **At risk** if $n_{\text{shrink}} > n_{\text{grow}}$ and $n_{\text{shrink}} \geq \lceil m/3 \rceil$.
3. **Growing** if $n_{\text{grow}} \geq \max(1,\ \mathrm{round}(0.4\,m))$ and τ_{all} is absent or ≥ 0.95.
4. **Holding** otherwise.

The "at risk" headline requires under-stimulated muscles to both outnumber growing ones and reach a third of those trained, so a few neglected small muscles do not flip the whole read. It resolves to "at risk" chiefly when most of recent training is genuinely sparse.

9 Plan forecast

Separately from the history-based signal, the plan forecast projects a plan's *prescribed* weekly sets before any of it is logged. Prescribed sets per rotation are scaled to a weekly rate by how often the plan is run, capped so an ambitious frequency on a short rotation cannot over-promise:

$$\hat{s}_g = s_g \cdot \min\left(1.2,\ \frac{p}{W}\right), \tag{14}$$

where s_g is prescribed sets per rotation (primary plus half of secondary, as in (1)), p is planned sessions per week, and W is the number of distinct rotation workouts. Each trained muscle is then sorted into under-dosed ($<$ MV), growth window (MEV to 20), past-productive ($>$ 20), or maintenance, and the counts drive a plain-language outlook. The forecast promises no centimetres or kilograms; it reports the dose against the growth window and what to change.

10 Scientific basis

Each parameter and mechanism traces to a specific source. Effect sizes are context-dependent; the model uses the literature to set structure and thresholds, not to promise outcomes. Grades are GRADE-style ratings of the underlying evidence: High (meta-analyses of RCTs), Moderate (individual RCTs or focused reviews), Low (practitioner framework). A Low grade flags a parameter that rests on weaker evidence and is the first candidate for revision.

Mechanism / parameter	**Basis**	**Grade**
Graded volume dose–response; $\sim$10 sets as a practical target	Schoenfeld, Ogborn & Krieger [9]: each added weekly set $\approx +0.37\%$ muscle, benefit rising across the studied range.	High
Fractional (0.5) counting of indirect sets; diminishing returns; frequency neutral at equal volume	Pelland et al. [5] (peer-reviewed; 67 studies, 2,058 subjects).	High
Continuous backbone $\rho(s)$ and landmark derivation	Derived from the dose–response curves above [9, 5].	High
Effort factor ε; hypertrophy comparable across $\sim$0–4 RIR, dropping off when easier	Refalo et al. [7], systematic review and meta-analysis of proximity-to-failure.	High
MEV/MAV landmark structure ($\sim$10 to 20 sets)	Israetel et al. [3], volume-landmarks framework.	Low
Maintenance floor MV; size held on $\sim$1/3 (older) to $\sim$1/9 (young) of building volume	Bickel, Cross & Bamman [1] (RCT, 16-wk build plus 32-wk reduced dose).	Moderate
Decline below maintenance stimulus; $\sim$2–4-week strength retention, then loss	Mujika & Padilla [4] detraining review; cross-sectional-area loss from $\sim$2–3 weeks of cessation.	Moderate
Overload via added load *or* reps, so the trend tracks both, not tonnage	Plotkin et al. [6] (RCT; load- and rep-progression gave near-identical growth); mechanism in Schoenfeld [8].	Moderate
Epley one-rep-max φ; self-calibrated denominator K	Standard load–rep relationship [2]; K fitted from the lifter's own Max sets (8).	Method

11 Verification

The model is not validated against body-composition measurement (see the limitations that follow), but three reproducible checks accompany it (the `research/` folder in the repository), each raising rigor without new measurement.

- **Known-answer tests.** Canonical scenarios — a returning lifter, a deload, easy versus hard sets, a hard stop — paired with domain-expected verdicts, so miscalibration or later drift is caught in code.

- **Sensitivity analysis.** Perturbing each threshold by $\pm 10\%$ over a synthetic population shows the growth-trend cutoff is the most load-bearing parameter (about a quarter of muscles change state), while the maintenance floor and sharp-drop cutoff are nearly cosmetic. This is why the borderline band sits on the trend boundary.

- **Predictive-coherence backtest.** On a training log, muscles labelled growing should precede larger subsequent gains in estimated 1RM than those labelled holding or under-stimulated. Strength change is a proxy for hypertrophy, not a direct measure; a real body-composition study remains the standard this cannot replace.

12 Assumptions and limitations

- **Validation status.** This is a mechanistic model grounded in the literature and internally consistent, but it has not been validated against direct body-composition measurements (DXA, ultrasound, or MRI). Its thresholds are derived from published dose–response findings, not fitted to an outcome dataset, and no predictive accuracy interval is claimed. The checks in Section 11 test internal consistency and coherence, not external accuracy.
- **Stimulus, not size.** Every output estimates the training stimulus from logs. The app cannot see the body; no measurement of muscle mass is implied.
- **Effort is self-reported.** Reps in reserve is not recoverable from load and reps alone; (6) sets only a default. Day-to-day max-rep variability (±1–3 reps) sits at the resolution of the effort bands, so the estimate is noisy near failure and the manual override remains the ground truth.
- **Top-set inference.** History stores the top set and a set count, not every set's reps, so within-session rep drop-off is not yet used as an effort cue.
- **One landmark set for all muscles.** Chosen for legibility; real landmarks vary by muscle and training age.
- **Load trend needs external load.** Bodyweight and timed work do not contribute to T^L; those muscles rely on the set trend alone.
- **Residual tonnage use.** The overall verdict's τ_{all} still uses whole-body volume; the per-muscle trend does not.
- **Conservative by design.** The maintenance floor and the "at risk" threshold are both set so the model under-reports muscle loss rather than over-reports it.

References

[1] Bickel CS, Cross JM, Bamman MM. Exercise dosing to retain resistance-training adaptations in young and older adults. *Med Sci Sports Exerc.* 2011;43(7):1177–1187. doi:10.1249/MSS.0b013e318207c15d

[2] Epley B. Poundage chart. *Boyd Epley Workout.* Lincoln, NE: Body Enterprises; 1985.

[3] Israetel M, Hoffmann J, Smith CW. Training volume landmarks for muscle growth (MEV/MAV/MRV). *Renaissance Periodization*; 2017. `https://rpstrength.com/blogs/articles/training-volume-landmarks-muscle-growth`

[4] Mujika I, Padilla S. Detraining: loss of training-induced physiological and performance adaptations. Part I — short-term insufficient training stimulus. *Sports Med.* 2000;30(2):79–87. doi:10.2165/00007256-200030020-00002

[5] Pelland JC, Remmert JF, Robinson ZP, Hinson SR, Zourdos MC. The resistance training dose response: meta-regressions exploring the effects of weekly volume and frequency on muscle hypertrophy and strength gains. *Sports Med.* 2026. doi:10.1007/s40279-025-02344-w

[6] Plotkin D, Coleman M, Van Every D, et al. Progressive overload without progressing load? The effects of load or repetition progression on muscular adaptations. *PeerJ.* 2022;10:e14142. doi:10.7717/peerj.14142

[7] Refalo MC, Helms ER, Trexler ET, Hamilton DL, Fyfe JJ. Influence of resistance training proximity-to-failure on skeletal muscle hypertrophy: a systematic review with meta-analysis. *Sports Med.* 2023;53(3):649–665. doi:10.1007/s40279-022-01784-y

[8] Schoenfeld BJ. The mechanisms of muscle hypertrophy and their application to resistance training. *J Strength Cond Res.* 2010;24(10):2857–2872. doi:10.1519/JSC.0b013e3181e840f3

[9] Schoenfeld BJ, Ogborn D, Krieger JW. Dose-response relationship between weekly resistance training volume and increases in muscle mass: a systematic review and meta-analysis. *J Sports Sci.* 2017;35(11):1073–1082. doi:10.1080/02640414.2016.1210197

A Symbol reference

Symbol	**Definition**
$g,\ x,\ e$	Muscle group, exercise, logged entry
$w,\ r,\ n,\ f$	Top-set load (kg), top-set reps, set count, effort code $\{0, 1, 2\}$
$\alpha(x, g)$	Muscle attribution weight: 1 primary, 0.5 secondary, 0 otherwise (1)
$\varepsilon(f)$	Effort factor: 0.5 / 1 / 1 for Easy / Hard / Max (2)
$S_{g,k}$	Effective weekly sets, muscle g, week k (3)
$\varphi(w, r),\ K$	Epley estimated one-rep max and its rep denominator (K=30 default, self-calibrated) (4), (8)
A_x	Anchor: best prior φ for exercise x (5)
$L_{g,k}$	Weekly peak φ on primary exercises for g
D_g	Recent dose: recent mean of S_g (11)
$\rho(s),\ s_0$	Dose–response stimulus fraction and its scale (9)
$T^S,\ T^L,\ \tau_g$	Set trend, load trend, combined trend (12)–(13)
σ_g	Per-muscle state (growing / holding / under-stimulated)
MV, MEV, target, MAV	Landmarks 2, 6, 10, 20 effective sets/muscle/week

B Parameter values

Every numeric constant in the model, its value, and where it enters. These are the values used in production (v121); the sensitivity analysis (Section 11) quantifies how much each one matters.

Parameter	Value	Where
Rolling window N	10 weeks	series length (3)
Recent / earlier windows	weeks 8–10 / weeks 5–7	(10)
Secondary-muscle weight	0.5	α (1)
Effort factors (Easy/Hard/Max)	0.5 / 1 / 1	ε (2)
RIR $\rightarrow$ effort bands	Max ≤ 1, Easy ≥ 4, else Hard	(7)
Epley denominator K	30 default, fitted then clamped to $[20, 40]$	$(4), (8)$
Dose–response scale s_0	target$/ \ln 5 \approx 6.21$	$\rho(s)$ (9)
Maintenance volume MV	2 effective sets/wk	state rule 1
Min. effective volume MEV	6 effective sets/wk	state rules 3–4
Practical target	10 effective sets/wk	radar goal; sets s_0
Upper bound MAV	20 effective sets/wk	plan forecast
Sharp-drop trend cutoff	$\tau_g < 0.75$	state rule 2
Growth trend cutoff	$\tau_g \geq 1.08$	state rule 3
Maintenance-band climb	$\tau_g \geq 1.15$ and $D_g \geq 5$	state rule 4
Borderline margins	± 1 set; trend within 0.06 of $\{0.75, 0.92, 1.08\}$	Section 7.1
Overall "at risk"	$n_{\text{shrink}} > n_{\text{grow}}$ and $n_{\text{shrink}} \geq \lceil m/3 \rceil$	Section 8
Overall "growing"	$n_{\text{grow}} \geq \max(1, \text{round}(0.4m))$, τ_{all} absent or ≥ 0.95	Section 8
"Need more data"	fewer than 3 active weeks	Section 8
Plan frequency cap	$\min(1.2,\ p/W)$	(14)

Yalla training-stimulus model, v121. Estimates a training stimulus from logs; not medical or diagnostic advice, and not a measurement of muscle mass.